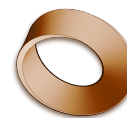


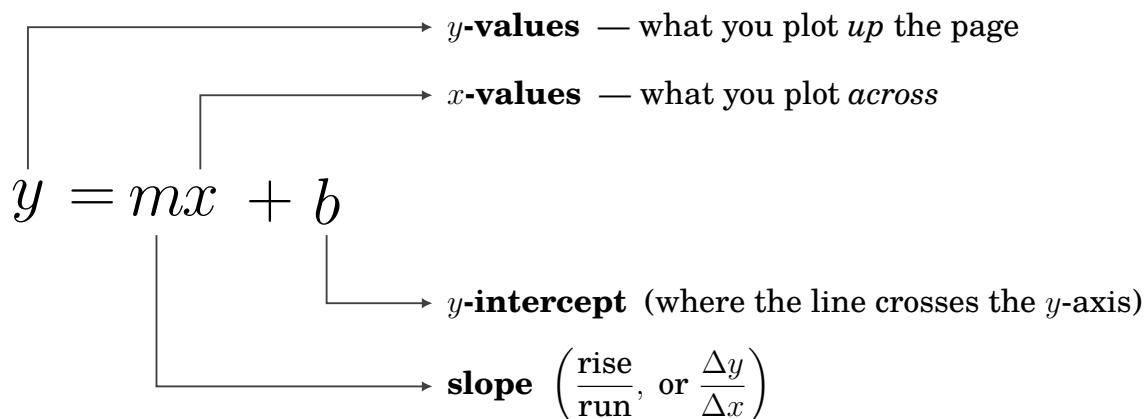
Name: \_\_\_\_\_  
 Class: \_\_\_\_\_  
 Date: \_\_\_\_\_



Mr. Rawson • Physics

## Straight Lines: $y = mx + b$ — ANSWERS

Every straight line that has ever been drawn can be written in one form. Learn the form once and it works in maths, in physics, and on every graph you will ever have to read the gradient of. **Only two numbers describe a straight line: how steep it is, and where it starts.**



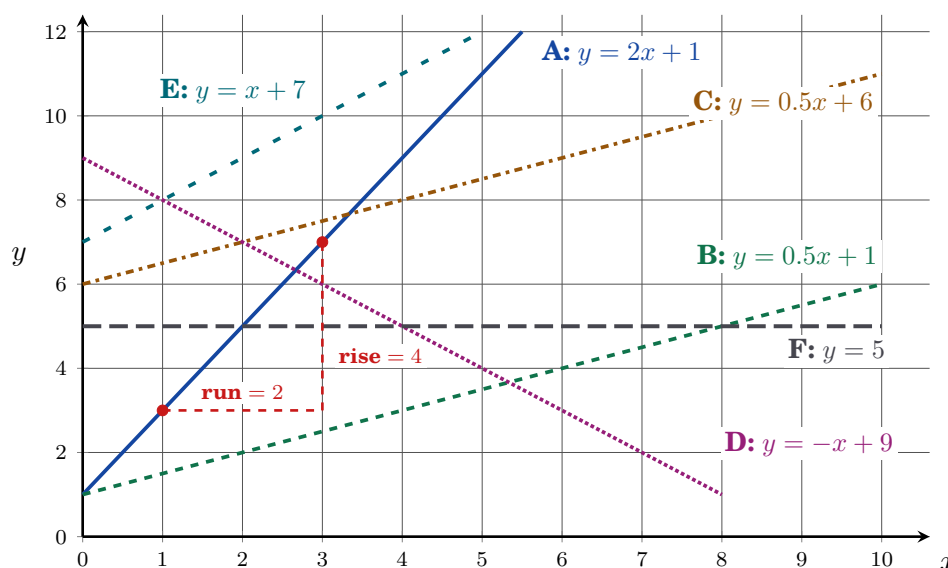
**The slope,  $m$ , is a rate: how much  $y$  changes for every 1 that  $x$  changes**

Pick **any two points** on the line. Go **across** from the first to the second — that is the **run**,  $\Delta x$ . Then go **up** — that is the **rise**,  $\Delta y$ . Divide:

$$m = \frac{\text{rise}}{\text{run}} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

$\Delta$  is the Greek letter *delta*, and in both maths and physics it always means “**the change in**”. It does *not* mean multiply.

Three things to hold on to. **It does not matter which two points you pick** — that is what makes a line straight. If the line goes **downhill** left to right the rise is negative, so  $m$  is negative. And a **flat** line has  $m = 0$ :  $y$  never changes, so the equation is just  $y = b$  — line **F** below is flat, and nowhere near zero.



**Worked on line A.** From (1, 3) to (3, 7): the run is  $3 - 1 = 2$  and the rise is  $7 - 3 = 4$ , so  $m = \frac{4}{2} = 2$ . Read straight off the equation  $y = 2x + 1$ :  $m = 2$ . They agree, because they are the same number said two ways.

1. Read  $m$  and  $b$  off the graph opposite

Line A is done for you. **You do not need the equations** — read  $b$  off where the line crosses the  $y$ -axis, and get  $m$  by drawing a triangle on the line, as in the worked example.

| Line | slope, $m$ | $y$ -intercept, $b$ | equation,<br>$y = mx + b$ |
|------|------------|---------------------|---------------------------|
| A    | 2          | 1                   | $y = 2x + 1$              |
| B    | 0.5        | 1                   | $y = 0.5x + 1$            |
| C    | 0.5        | 6                   | $y = 0.5x + 6$            |
| D    | -1         | 9                   | $y = -x + 9$              |
| E    | 1          | 7                   | $y = x + 7$               |
| F    | 0          | 5                   | $y = 5$                   |

## 2. Now compare them

(a) Two of the lines are **parallel**. Which two, and what do their equations have in common?

**B and C. They have the same  $m$  (0.5), so they are equally steep — and lines with the same slope never meet. Only their  $b$  differs, which slides the line up the page without tilting it.**

(b) Two of the lines cross the  $y$ -axis at the **same point**. Which two, and what does that tell you about  $b$ ?

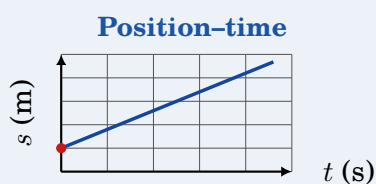
**A and B. Both have  $b = 1$ , so both start at  $(0, 1)$  — but they have different slopes, so they immediately separate.  $b$  fixes where a line starts;  $m$  fixes where it goes.**

(c) How can you tell that line D has a negative slope *before* doing any arithmetic at all?

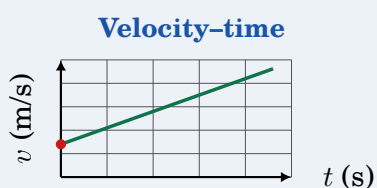
**It goes downhill as you read left to right. As  $x$  gets bigger,  $y$  gets smaller, so the rise is negative and  $m$  must be too.**

## Why physics cares — the three graphs of motion

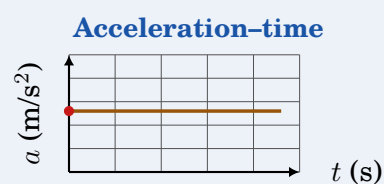
On a physics graph the axes are named quantities, not  $y$  and  $x$  — but you read them the same way. Three graphs describe any motion, and **the gradient means something different on each**. The red dot is the intercept.



gradient = **velocity**,  $v$   
intercept = where it **started**



gradient = **acceleration**,  $a$   
intercept = **starting velocity**,  $u$



gradient = **zero** — it is flat  
its **height** is the acceleration

**Now read them across, left to right.** The gradient of the first graph is the quantity plotted up the second; the gradient of the second is the quantity plotted up the third. *That is the whole chain*, and it is why **a gradient is never just a number — it is a different quantity, with a different unit.**

**Try it.** A trolley's **velocity-time** graph is a straight line crossing the vertical axis at  $3 \text{ m/s}$  and climbing  $2 \text{ m/s}$  for every  $1 \text{ s}$  across.

(a) The gradient, with its unit, and its name?  $2 \text{ m/s}^2$  — **the acceleration**. (b) So its **acceleration-time** graph looks like? **A flat line at  $2 \text{ m/s}^2$ .**